VerdantAbode, CO_2 Control System

User Manual

This document describes how to use the available interfaces on the device to configure and use its functionality.

Local Interface

The device has a screen with two distinct views: the default system view and the wifi setup view. Both views are operated using the three buttons and the rotary encoder.

System View

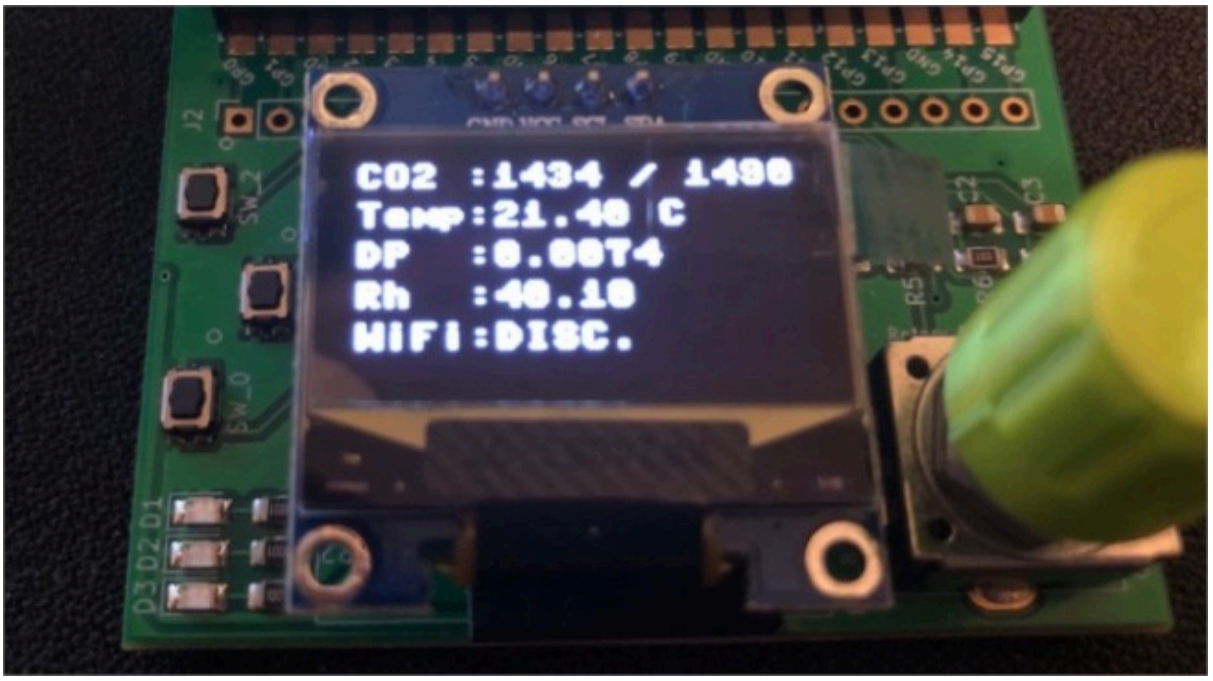

Figure 1: The main system view displaying environmental values and network status.

Changing the CO_2 target level is done by rotating the rotary encoder when in system view. The range is limited between 200 and 1500ppm. Saving the device state, including the CO_2 target level, can be done manually by pressing *SW_2*. The Wi-Fi value will change to reflect current system network status (disconnected or connected).

Wi-Fi Setup View

Toggling to the Wi-Fi setup screen from the default view is done by pressing *SW_0*.

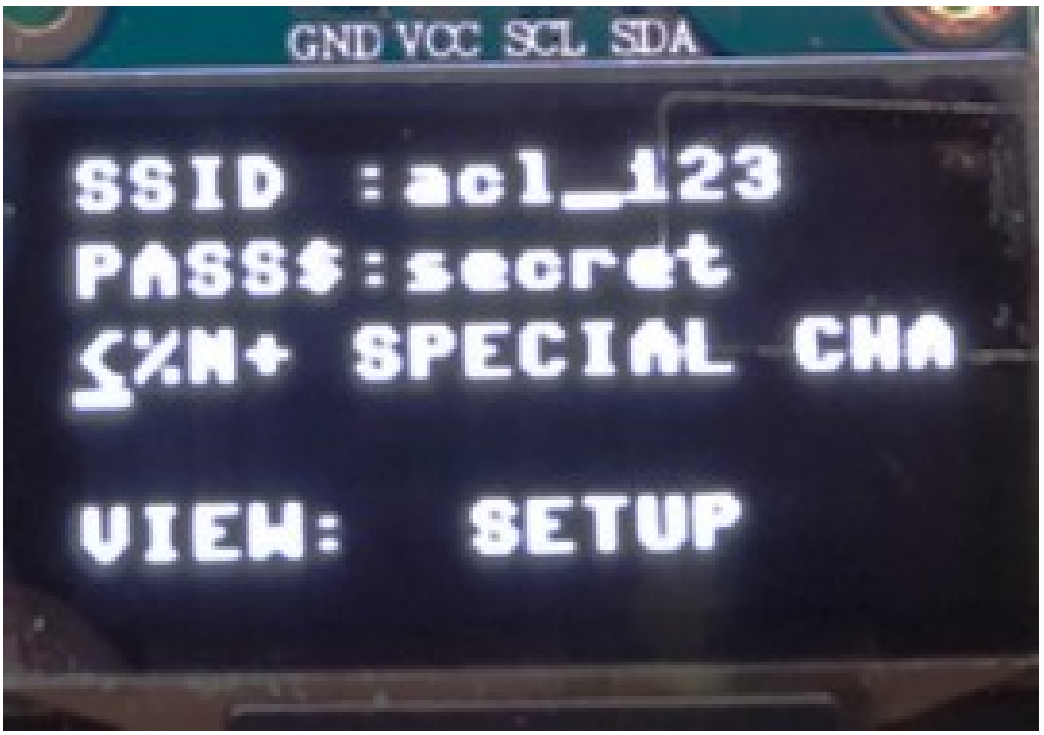

Figure 2: Example view of setting SSID details within Wi-Fi setup.

The Wi-Fi setup view consists of three fields: SSID, PASS, and character selection. The currently active field (SSID/PASS) is denoted by the '$' symbol before the colon. The colon is followed by the currently inputted characters into the field.

Inserting characters is done by rotating the rotary encoder to the desired character (denoted by '_'), and pressing *SW_1* to insert it to the end of the current field.

The available characters are split by category (a-z, A-Z, 0-9, and symbols), which can be rotated through by pressing *SW_2*. There is also a special characters section (see figure 2), which has special functionality mapped to each character. See Table 1 for the mappings.

<	Delete latest character in active field
%	Clear active field entirely
N	Move to next field
+	Connect to network

Table 1: Wi-Fi setup special character's mappings.

Insert the desired character, or execute special actions, by pressing *SW_1* when the correct character is highlighted with the '_'. Finish by rotating to the special characters lineup and press '+' to attempt a connection with entered credentials.

ThingSpeak (Logging Data)

When the system is connected to a Wi-Fi network it will periodically send data to the configured ThingSpeak backend (30 second interval by default). If the Wi-Fi link goes down, the system will attempt to re-connect using the existing credentials periodically.

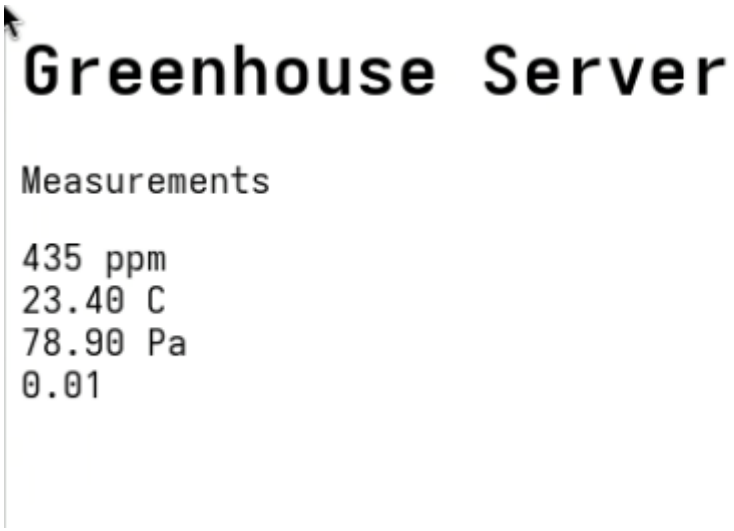

Figure 3: Sensor data sent to ThingSpeak.

ThingSpeak (Receiving Commands)

Feature not implemented.